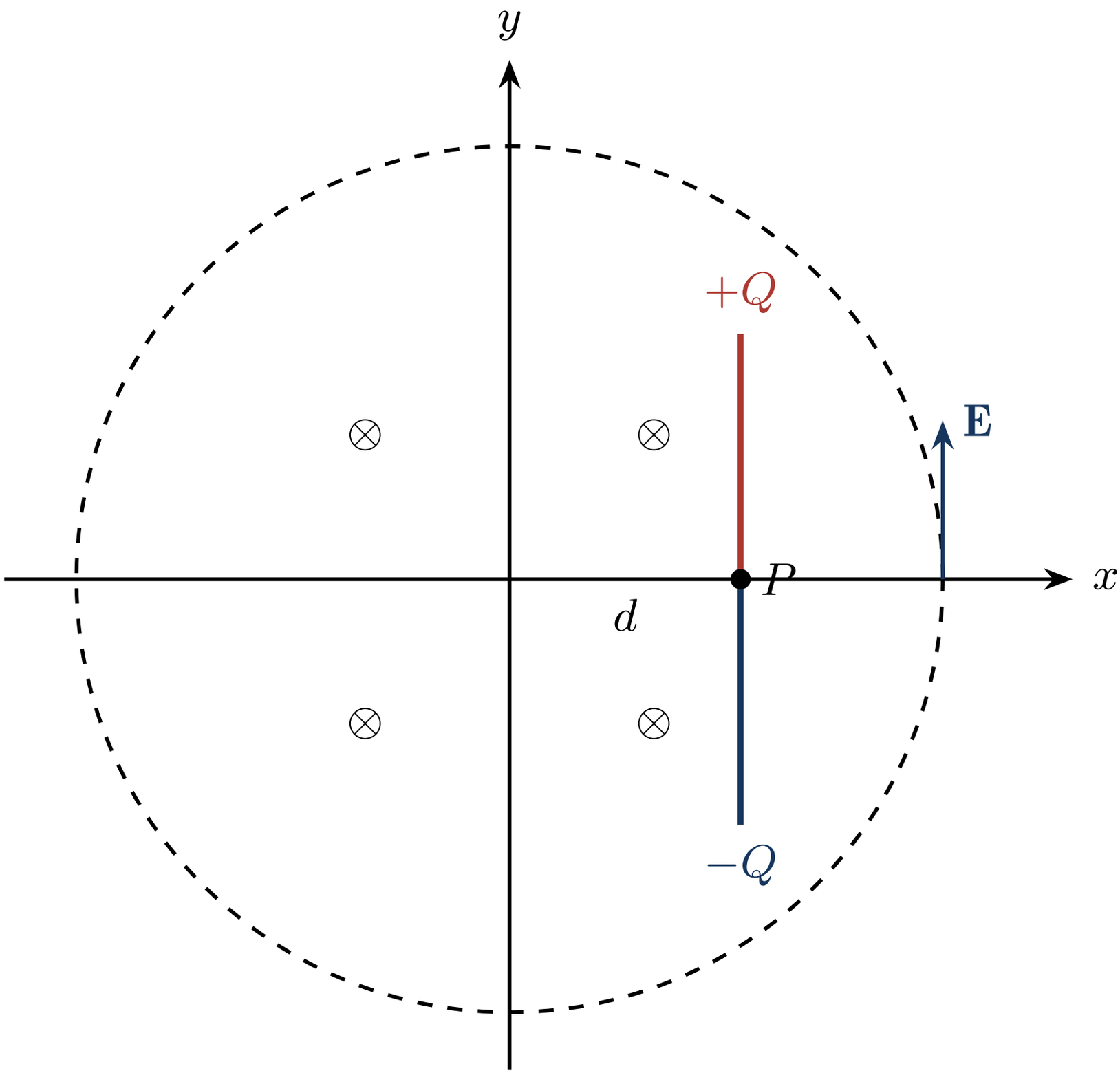
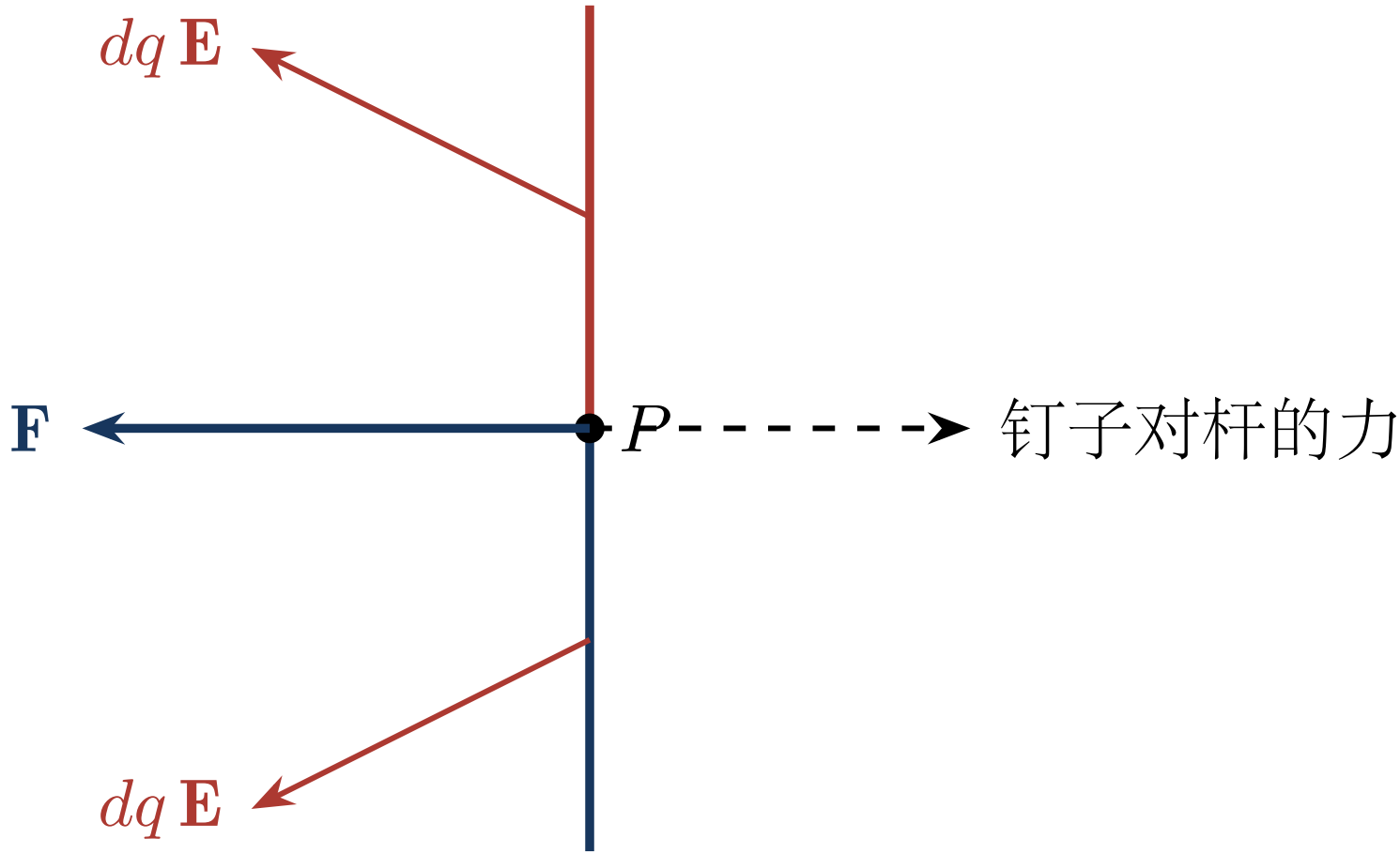
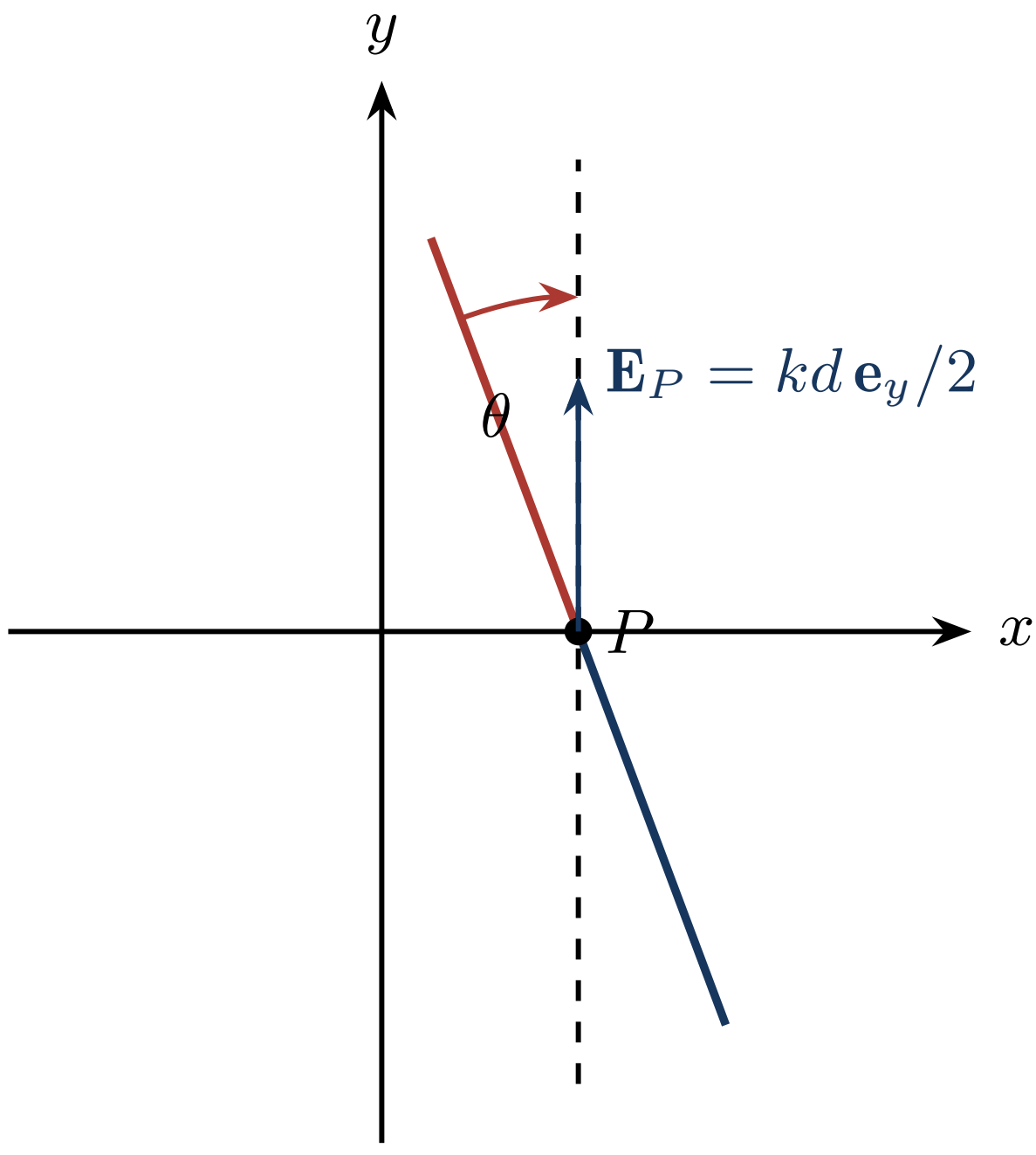


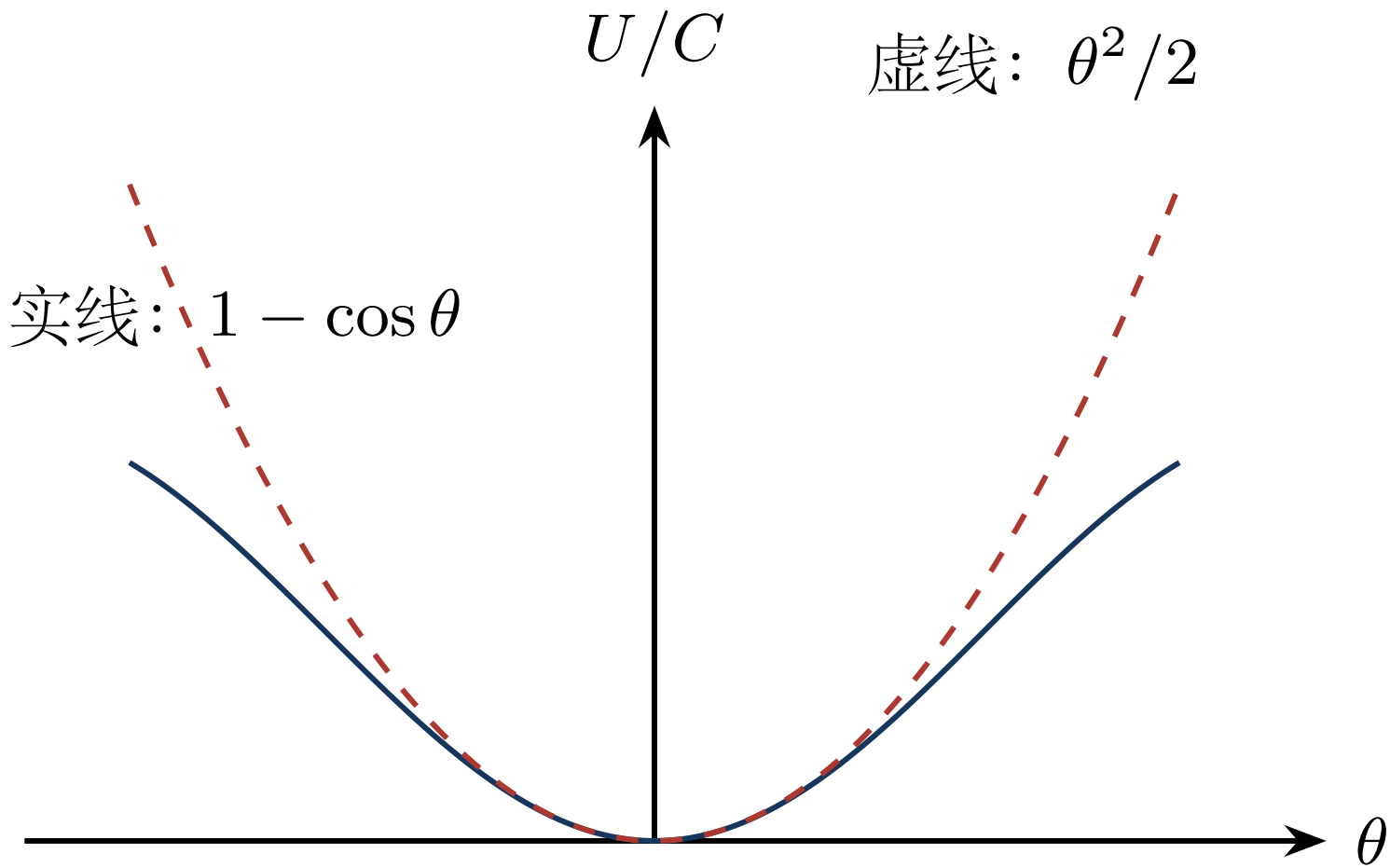
磁场朝纸内且以速率 k 增强；感生电场绕圆心逆时针。



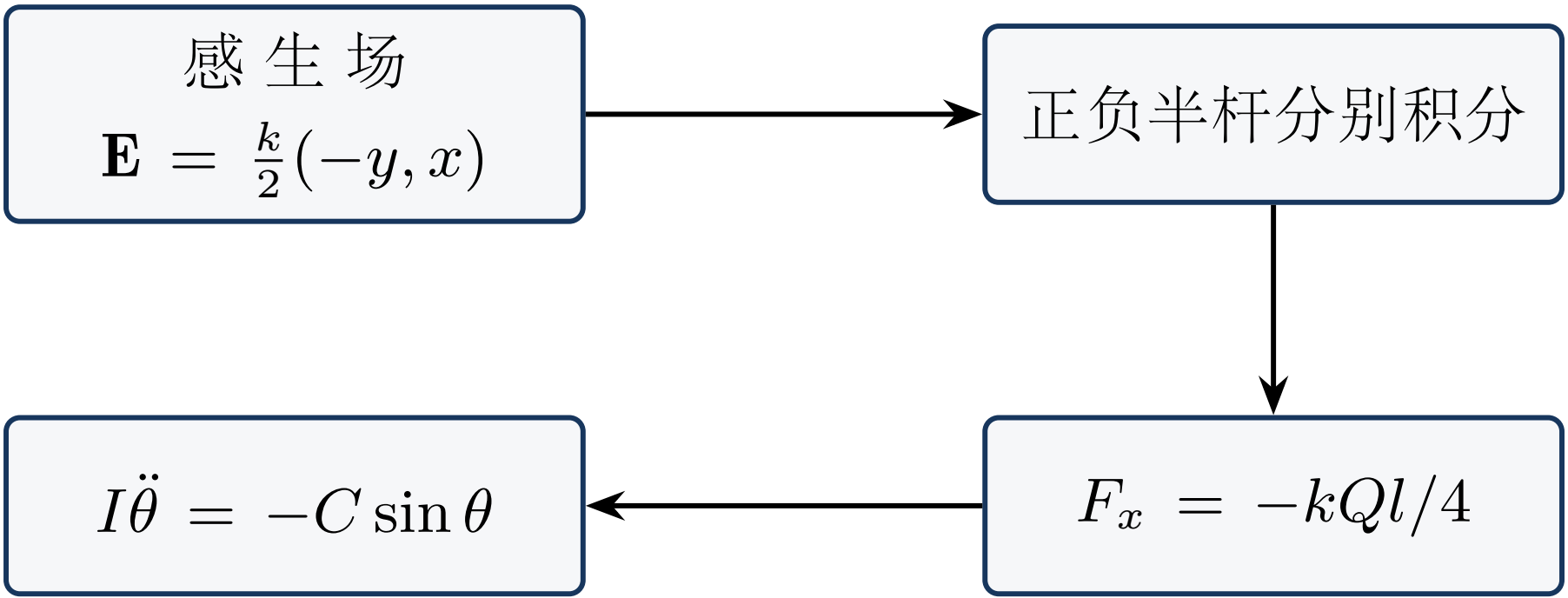
上下半杆的竖直分力抵消，水平分力相加。杆对钉子的初始力向左，大小 $kQl/4$。



上端向左偏离竖直时，$\tau_P = -kQld\sin\theta/4$，力矩使杆回到竖直。



$C = kQld/4$；此势能只针对杆的单一转动自由度。小振动 $\omega^2 = 3kQd/(ml)$。



先判断力和力矩方向，再线性化；小振幅周期 $2\pi\sqrt{ml/(3kQd)}$。